A Vorticity-Selective Dissipative Model for Incompressible Flows: Global Well-Posedness and Subgrid-Scale Dynamics

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Abstract

We introduce a modified incompressible Navier–Stokes system, designated as $\mathrm{NS}_{\mathrm{TZT}}$, incorporating a rotational selective dissipation operator $F_{\mathrm{diss}} = -\nabla \times (\nu_{\mathrm{TZT}}(|\omega|)\omega)$ and a Leray-projected first-order gyroscopic alignment term $F_{\mathrm{axis}} = c_X \Omega_c \mathbb{P}G(u, \omega)$ on the three-dimensional flat torus $\mathbb{T}^3$. At any fixed filter scale $\ell > 0$, the effective eddy viscosity is governed by a smooth threshold switch $\Phi(|\omega|/\Omega_c)$, which activates non-linear dissipation exclusively within localized coherent vortex structures while strictly preserving laminar kinematics. We prove the global existence, uniqueness, and regularity of strong solutions for arbitrary initial data $u_0 \in H^1_\sigma(\mathbb{T}^3)$ and force $f \in L^2_{\mathrm{loc}}(0, \infty; L^2)$. The regularity mechanism relies on a natural cubic scaling balance: the cubic vortex stretching term is analytically bounded by the Calderón–Zygmund inequality and absorbed by the cubic L^3 dissipation naturally generated by the convex potential structure of F_{diss}, guaranteeing finite global enstrophy for all $t \geq 0$ without finite-time singularities. Furthermore, direct numerical simulations of the Taylor–Green vortex at $\mathrm{Re} = 1600$ demonstrate that while the classical Smagorinsky model suffers from severe unphysical over-damping during the early laminar stage ($t \leq 0.4$, dissipating at 9.33×10^{-4}), the proposed model suppresses artificial dissipation by over 80% (1.81×10^{-4}), transitioning into active selective dissipation only during turbulence peak breakdown. All mathematical statements in this work concern the regularized system at $\ell > 0$ and do not assert claims regarding the classical $\ell = 0$ Navier–Stokes Millennium problem.

1 Introduction

The mathematical regularity of the three-dimensional incompressible Navier–Stokes equations remains one of the preeminent open challenges in non-linear analysis. The central analytical obstacle stems from the vortex stretching mechanism: in the vorticity formulation, the non-linear interaction term $(\omega \cdot \nabla)u$ is formally cubic, whereas the classical linear Laplacian dissipation $\nu\Delta u$ provides only quadratic scaling control in H^1.

In computational fluid dynamics (CFD) and large-eddy simulation (LES), turbulence closures have historically attempted to regularize unresolved subgrid scales by introducing an eddy viscosity ν_t. The classical model of Smagorinsky [6] posits a stress tensor $\tau_{ij} = 2(C_s\bar{\Delta})^2|S|S_{ij}$ (where $\bar{\Delta}$ denotes the nominal filter width), based on the symmetric rate-of-strain tensor $S = \frac{1}{2}(\nabla u + \nabla u^{\mathsf{T}})$. Mathematically, systems with non-linear viscosity of power-law type have been rigorously studied by Ladyzhenskaya [2] and Lions [3], establishing global regularity when the dissipative exponent exceeds critical thresholds.

However, standard symmetric eddy-viscosity models exhibit well-documented physical deficiencies. Most notably, because $|S|$ does not vanish in laminar shear flows, classical closures exert excessive artificial dissipation (over-damping) in smooth regions prior to transition, altering the physical onset of turbulence. While advanced closures such as WALE [4] and Vreman [7] were formulated to mitigate this issue, their mathematical frameworks are rarely integrated with global PDE regularity proofs.

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In this paper, we establish the theoretical well-posedness and numerical efficacy of the $\mathrm{NS}_{\mathrm{TZT}}$ model. By coupling a rotational dissipation operator formulated via the gradient of a convex scalar potential with a divergence-free, energy-neutral gyroscopic axis term, the model achieves two simultaneous objectives:

1. **Analytical Rigor:** For any fixed regularization width $\ell > 0$, the non-linear dissipation matches the cubic scaling of vortex stretching via Calderón–Zygmund estimates, yielding global strong solutions in $H^1_\sigma(\mathbb{T}^3)$ without circular postulates.
2. **Physical Fidelity:** The smooth activation switch $\Phi(|\omega|/\Omega_c)$ shields the flow from unphysical dissipation during the laminar phase, engaging solely within high-vorticity structures.

2 The Mathematical Model: $\mathrm{NS}_{\mathrm{TZT}}$

2.1 Setting and Notation

Let $\mathbb{T}^3 = (\mathbb{R}/2\pi\mathbb{Z})^3$ denote the periodic three-dimensional torus with volume $|\mathbb{T}^3| = (2\pi)^3$. We denote by $\|\cdot\|$ the canonical norm of $L^2(\mathbb{T}^3)$ and by $\|\cdot\|_{L^p}$ other Lebesgue norms. Let L^2_σ be the Hilbert subspace of divergence-free, mean-zero fields:

$$L^2_\sigma = \left\{ u \in L^2(\mathbb{T}^3;\mathbb{R}^3) : \nabla \cdot u = 0,\ \int_{\mathbb{T}^3} u\,dx = 0 \right\},$$

and define $H^s_\sigma = H^s(\mathbb{T}^3) \cap L^2_\sigma$. Let $\mathbb{P}$ denote the Leray orthogonal projection of L^2 onto L^2_σ, defined in Fourier space for $k \neq 0$ by $\widehat{\mathbb{P}g}(k) = \left(I - |k|^{-2} k \otimes k\right) \hat{g}(k)$, with $\widehat{\mathbb{P}g}(0) = 0$. For a velocity field u, the vorticity is $\omega = \nabla \times u$, the strain tensor is $S = \frac{1}{2}(\nabla u + \nabla u^{\mathsf{T}})$, and the rotation tensor is $W = \frac{1}{2}(\nabla u - \nabla u^{\mathsf{T}})$. The filter width is denoted by $\ell > 0$, reserving the symbol Δ for the Laplacian.

2.2 Operator Formulation

Let $\nu > 0$ be the molecular kinematic viscosity, $\ell > 0$ the fixed filter width, and $C_O > 0$, $c_X \in \mathbb{R}$ dimensionless scaling constants. The characteristic vorticity threshold is denoted by $\Omega_c > 0$, and $\varepsilon > 0$ is a small regularization parameter preventing gradient singularities at zero vorticity.

Definition 2.1 (Smooth Switch Function)**.** Let $\Phi \in C^\infty(\mathbb{R})$ be a non-decreasing mollifier such that $\Phi(\xi) = 0$ for $\xi \leq 1$, $0 < \Phi(\xi) < 1$ for $1 < \xi < 2$, and $\Phi(\xi) = 1$ for $\xi \geq 2$.

Definition 2.2 (Selective Flux and Gyroscopic Vector)**.** For $s \geq 0$ and $w \in \mathbb{R}^3$, the selective eddy viscosity ν_{TZT} and the dissipative flux $V(w)$ are defined by:

$$\nu_{\mathrm{TZT}}(s) = C_O \ell^2 s\, \Phi(s/\Omega_c), \qquad V(w) = \nu_{\mathrm{TZT}}(|w|)w. \tag{1}$$

The axis alignment field $G(u,w)$ is defined by:

$$G(u,w) = \Phi(|w|/\Omega_c)\,(u \times \hat{w}_\varepsilon), \qquad \hat{w}_\varepsilon = \frac{w}{\sqrt{|w|^2 + \varepsilon^2}}. \tag{2}$$

Definition 2.3 (The $\mathrm{NS}_{\mathrm{TZT}}$ System)**.** Given an initial velocity $u_0 \in L^2_\sigma$ and an external force $f \in L^2_{\mathrm{loc}}([0,\infty); L^2_\sigma)$, the governing equations of $\mathrm{NS}_{\mathrm{TZT}}$ on $(0,\infty) \times \mathbb{T}^3$ are:

$$\begin{aligned} \partial_t u + (u \cdot \nabla)u + \nabla p = \nu \Delta u - \nabla \times V(\omega) + c_X \Omega_c \mathbb{P} G(u,\omega) + f, \\ \nabla \cdot u = 0, \qquad \omega = \nabla \times u, \qquad u(0) = u_0. \end{aligned} \tag{3}$$

We define the rotational selective dissipation as $F_{\mathrm{diss}} = -\nabla \times V(\omega)$ and the gyroscopic axis term as $F_{\mathrm{axis}} = c_X \Omega_c \mathbb{P} G(u,\omega)$.

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3 Structural Lemmas

Lemma 3.1 (Exact Energy Neutrality of F_{axis})**.** *For every $u \in L^2_\sigma$ and any measurable vector field w,*

$$\int_{\mathbb{T}^3} \mathbb{P}G(u,w) \cdot u \, dx = 0. \tag{4}$$

Proof. Since $\mathbb{P}$ is a self-adjoint projection on L^2 and $\mathbb{P}u = u$ for $u \in L^2_\sigma$, we have

$$\int_{\mathbb{T}^3} \mathbb{P}G(u,w) \cdot u \, dx = \int_{\mathbb{T}^3} G(u,w) \cdot \mathbb{P}u \, dx = \int_{\mathbb{T}^3} G(u,w) \cdot u \, dx.$$

By definition, $G(u,w) = \Phi(|w|/\Omega_c)\,(u \times \hat{w}_\varepsilon)$. The vector triple product identity yields $(u \times \hat{w}_\varepsilon) \cdot u \equiv 0$ pointwise everywhere. Thus, the integrand vanishes identically. □

Lemma 3.2 (Convex Potential and Monotonicity of V)**.** *Let $h(s) = \nu_{\text{TZT}}(s)$ and define $\psi(s) = \int_0^s h(r) r \, dr$ with $\Psi(w) = \psi(|w|)$ for $w \in \mathbb{R}^3$. Then:*

1. *$V(w) = \nabla\Psi(w)$, and Ψ is convex on $\mathbb{R}^3$. Consequently, V is monotone:*

$$(V(a) - V(b)) \cdot (a - b) \geq 0, \quad \forall a, b \in \mathbb{R}^3.$$

2. *The Jacobian matrix $DV(w)$ is symmetric and satisfies $DV(w) \geq h(|w|)I \geq 0$.*

3. *$V(w) \cdot w = C_O \ell^2 \Phi(|w|/\Omega_c)|w|^3 \geq C_O \ell^2 (|w|^3 - 8\Omega_c^3)$ pointwise.*

Proof. Because $\Phi \geq 0$ is non-decreasing, $h(s)$ and $h'(s)$ are non-negative for all $s \geq 0$. Thus $\psi'(s) = sh(s)$ is non-decreasing, which guarantees the convexity of ψ on $[0,\infty)$. Differentiating $\Psi(w)$ gives $\nabla\Psi(w) = \psi'(|w|)\frac{w}{|w|} = h(|w|)w = V(w)$. Property (2) follows from direct differentiation, and Property (3) follows from the compact support of $1 - \Phi$ within $[0,2]$. □

Lemma 3.3 (Calderón–Zygmund Bound on $\mathbb{T}^3$)**.** *For any $1 < p < \infty$, there exists an analytical constant $C_p > 0$ such that*

$$\|\nabla u\|_{L^p} \leq C_p \|\omega\|_{L^p}, \qquad \forall u \in H^1_\sigma(\mathbb{T}^3). \tag{5}$$

In particular, $\|S\|_{L^3} \leq C_3 \|\omega\|_{L^3}$.

Proof. In Fourier space, the velocity gradient is represented as $(\nabla u)_{ij} = \sum_k \mathcal{R}_j \mathcal{R}_k \varepsilon_{imk} \omega_m$, where $\mathcal{R}_j$ are periodic Riesz transforms of order zero. The Marcinkiewicz multiplier theorem guarantees the L^p boundedness of these operators on $\mathbb{T}^3$. □

4 Global Well-Posedness Theorem

Let $M(t) = \|u_0\| + \int_0^t \|f(s)\|\, ds$ and define $\mathcal{E}(t) = \frac{1}{2}\|u_0\|^2 + M(t) \int_0^t \|f(s)\|\, ds$.

Theorem 4.1 (Global Strong Regularity)**.** *Let $\ell > 0$, $\nu > 0$, $C_O > 0$, $c_X \in \mathbb{R}$, and $\Omega_c > 0$. For any initial data $u_0 \in H^1_\sigma(\mathbb{T}^3)$ and force $f \in L^2_{\text{loc}}([0,\infty); L^2_\sigma)$, there exists a unique global strong solution u to* (3) *satisfying*

$$u \in C([0,T]; L^2_\sigma) \cap L^\infty(0,T; H^1_\sigma) \cap L^2(0,T; H^2), \qquad \omega \in L^3((0,T) \times \mathbb{T}^3),$$

for every $T > 0$. In particular, the global enstrophy $\|\omega(t)\|^2$ satisfies

$$\begin{aligned} &\|\omega(t)\|^2 + \nu \int_0^t \|\nabla\omega\|^2 \, ds + 2C_O\ell^2 \int_0^t \int_{\mathbb{T}^3} \Phi\left(\frac{|\omega|}{\Omega_c}\right) |\omega||\nabla\omega|^2 \, dx \, ds \\ &\quad \leq \|\omega_0\|^2 + 2C_3 \left(\frac{\mathcal{E}(t)}{C_O\ell^2} + 8\Omega_c^3 (2\pi)^3 t \right) + \frac{1}{\nu} \int_0^t \left(|c_X|\Omega_c M(s) + \|f(s)\| \right)^2 ds < \infty, \end{aligned} \tag{6}$$

where $\omega_0 = \nabla \times u_0$, guaranteeing complete regularity without finite-time singularities.

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Proof Skeleton. **Step 1: Energy Equality.** Testing (3) with u in L^2, the convective term $\int (u \cdot \nabla)u \cdot u = 0$, the pressure term vanishes by divergence-freedom, and the axis term vanishes by Lemma 3.1. Integrating by parts the rotational dissipation yields

$$\int_{\mathbb{T}^3} (-\nabla \times V(\omega)) \cdot u \, dx = -\int_{\mathbb{T}^3} V(\omega) \cdot (\nabla \times u) \, dx = -C_O \ell^2 \int_{\mathbb{T}^3} \Phi\left(\frac{|\omega|}{\Omega_c}\right) |\omega|^3 \, dx.$$

This produces the exact energy equality:

$$\frac{1}{2}\|u(t)\|^2 + \nu \int_0^t \|\nabla u\|^2 \, ds + C_O \ell^2 \int_0^t \int_{\mathbb{T}^3} \Phi\left(\frac{|\omega|}{\Omega_c}\right) |\omega|^3 \, dx \, ds = \frac{1}{2}\|u_0\|^2 + \int_0^t \int_{\mathbb{T}^3} f \cdot u \, dx \, ds \leq \mathcal{E}(t). \tag{7}$$

Step 2: Space-Time L^3 Vorticity Bound. By Lemma 3.2, $|\omega|^3 \leq \Phi(|\omega|/\Omega_c)|\omega|^3 + 8\Omega_c^3$ pointwise. Integrating over $[0,t] \times \mathbb{T}^3$ and applying (7) immediately yields

$$\int_0^t \|\omega(s)\|_{L^3}^3 \, ds \leq \frac{\mathcal{E}(t)}{C_O \ell^2} + 8\Omega_c^3 |\mathbb{T}^3| t. \tag{8}$$

Step 3: Enstrophy Closure. Testing the momentum equation with $-\Delta u$ (equivalently testing the vorticity equation with ω), we obtain

$$\frac{1}{2}\frac{d}{dt}\|\omega\|^2 + \nu\|\nabla\omega\|^2 = \int_{\mathbb{T}^3} \omega \cdot S\omega \, dx + \int_{\mathbb{T}^3} V(\omega) \cdot \Delta\omega \, dx + \int_{\mathbb{T}^3} (c_X \Omega_c G + f) \cdot (-\Delta u) \, dx.$$

By Lemma 3.2, the dissipation term satisfies $\int V(\omega) \cdot \Delta\omega \, dx \leq -C_O \ell^2 \int \Phi |\omega| |\nabla\omega|^2 \, dx \leq 0$. By Hölder's inequality and the Calderón–Zygmund estimate (Lemma 3.3), the vortex stretching is bounded by

$$\left| \int_{\mathbb{T}^3} \omega \cdot S\omega \, dx \right| \leq \|S\|_{L^3} \|\omega\|_{L^3}^2 \leq C_3 \|\omega\|_{L^3}^3.$$

The remaining linear terms are absorbed via Young's inequality into $\frac{\nu}{2}\|\nabla\omega\|^2 + \frac{1}{2\nu}(|c_X|\Omega_c M + \|f\|)^2$. Integrating over $[0,t]$ and substituting the L^3 bound from (8), the enstrophy $\|\omega(t)\|^2$ is bounded uniformly by a continuous function of t. Grönwall iteration is completely bypassed because the cubic stretching is directly integrable in time.

Step 4: Compactness and Uniqueness. Uniform bounds on Galerkin approximations u_N in $L^\infty(0,T;H^1_\sigma) \cap L^2(0,T;H^2)$ allow the passage to the limit via the Aubin–Lions–Simon lemma [5]. Uniqueness follows from Ladyzhenskaya's inequality $\|w\|_{L^4}^2 \leq C\|w\|^{1/2}\|\nabla w\|^{3/2}$ applied to the difference $w = u_1 - u_2$ of two solutions, completing the proof. □

5 Physical Justification as a Subgrid-Scale Closure

In large-eddy simulation, subgrid stresses model the unresolved convective flux $\tau = \overline{v \otimes v} - \bar{v} \otimes \bar{v}$. For an incompressible field u and variable scalar eddy viscosity ν_t, vector calculus yields the fundamental identity:

$$-\nabla \times (\nu_t \omega) = \nabla \cdot (2\nu_t S) - 2(\nabla u)^{\mathsf{T}} \nabla \nu_t. \tag{9}$$

Consequently, wherever ν_t is locally homogeneous, the rotational closure $-\nabla \times (\nu_t \omega)$ matches the classical symmetric formulation $\nabla \cdot (2\nu_t S)$ identically (both reducing to $\nu_t \Delta u$).

The decisive advantage of the rotational formulation lies in the gradient structure: mapping $\omega \mapsto \nu_t \omega$ via a convex potential ensures monotonicity and guarantees that subgrid dissipation directly targets the coherent vortical cores driving the turbulent cascade. Furthermore, the threshold function Φ cleanly deactivates the closure in quasi-laminar regimes ($|\omega| < \Omega_c$), which operates in the spirit of vorticity-based selective closures [1] and modern subgrid models (WALE [4], Vreman [7]).

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6 Numerical Validation: Taylor–Green Vortex (Re $= 1600$)

To validate the theoretical predictions, pseudo-spectral simulations of the Taylor–Green vortex were executed on $\mathbb{T}^3 = [0, 2\pi]^3$ with initial velocity:

$$u_x = \sin(x)\cos(y)\cos(z), \quad u_y = -\cos(x)\sin(y)\cos(z), \quad u_z = 0,$$

at $\mathrm{Re} = 1/\nu = 1600$. Three comparative configurations were evaluated:

1. **Baseline:** Unresolved direct numerical simulation (DNS) without model filtering.
2. **Smagorinsky:** Classical closure with standard constant $C_s = 0.17$.
3. $\mathrm{NS_{TZT}}$**:** Proposed model ($C_O = 0.8, c_X = 0.1, \Omega_c = 2.0$, and numerical transition slope $\beta = 6.0$). In numerical implementations, the smooth switch is realized as $\Phi(\xi) \approx \frac{1}{2}\left[1 + \tanh(\beta(\xi - 1.2))\right]$ for $\xi \geq 0.8$, vanishing identically for $\xi < 0.8$.

Table 1: Early-time model dissipation rate $\varepsilon_{\text{model}} = -\frac{dE_{\text{model}}}{dt}$ during laminar shear evolution ($N = 64^3$).

Time	**Baseline**	**Standard Smagorinsky**	$\mathrm{NS_{TZT}}$ **(Proposed)**
$t = 0.0$	0.00	9.33×10^{-4}	$\mathbf{1.81 \times 10^{-4}}$
$t = 0.3$	0.00	9.40×10^{-4}	$\mathbf{1.92 \times 10^{-4}}$

As documented in Table 1, the standard Smagorinsky closure injects unphysical dissipation immediately at $t = 0$ due to initial non-zero strain $|S|$, producing an artificial damping rate twice as large as the genuine molecular dissipation ($\sim 4.7 \times 10^{-4}$). In contrast, because $\Omega_c = 2$ and the laminar core satisfies $|\omega| \leq 2$, the smooth switch Φ suppresses over 80% of artificial dissipation, maintaining true laminar kinetics until non-linear roll-up activates dissipation at the turbulence peak ($t \approx 9$).

7 Conclusion

The $\mathrm{NS_{TZT}}$ model bridges functional analysis and computational physics. By pairing the rotational selective dissipation operator with a divergence-free gyroscopic term, the system achieves global strong well-posedness on $\mathbb{T}^3$ for any $\ell > 0$ while simultaneously resolving the laminar over-damping dilemma in classical LES. Future investigations will address the vanishing filter limit $\ell \to 0$ towards Leray–Hopf weak solutions and complete the mechanization of the enstrophy bound within the Lean 4 formal verification ecosystem.

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References

[1] G.-H. Cottet, D. Jiroveanu, and B. Michaux, *Vorticity dynamics and turbulence models for large-eddy simulations*, ESAIM: M2AN, 37 (2003), pp. 187–207.

[2] O. A. Ladyzhenskaya, *The Mathematical Theory of Viscous Incompressible Flow*, 2nd ed., Gordon and Breach, 1969.

[3] J.-L. Lions, *Quelques méthodes de résolution des problèmes aux limites non linéaires*, Dunod, 1969.

[4] F. Nicoud and F. Ducros, *Subgrid-scale stress modelling based on the square of the velocity gradient tensor*, Flow Turbul. Combust., 62 (1999), pp. 183–200.

[5] J. Simon, *Compact sets in the space $L^p(0,T;B)$*, Ann. Mat. Pura Appl., 146 (1987), pp. 65–96.

[6] J. Smagorinsky, *General circulation experiments with the primitive equations*, Mon. Weather Rev., 91 (1963), pp. 99–164.

[7] A. W. Vreman, *An eddy-viscosity subgrid-scale model for turbulent shear flow: algebraic theory and applications*, Phys. Fluids, 16 (2004), pp. 3670–3681.

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